

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 670 Information Retrieval — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Manning, Raghavan & Schütze — Introduction to Information Retrieval (free)	The primary text, free in full from the authors. Modules 02 through 06 and 10 follow it closely, and its treatment of the index and the scoring derivations is still unmatched.
Croft, Metzler & Strohman — Search Engines: Information Retrieval in Practice (free PDF)	The systems companion, also free. Stronger on Modules 04, 09, and 10 — crawling, query processing, and what a production engine actually contains.
Lin, Nogueira & Yates — Pretrained Transformers for Text Ranking (free)	Modules 06, 07, and 11, free. The best available survey of the neural ranking material, and honest about which gains survived scrutiny.
Zobel & Moffat — Inverted files for text search engines (free)	Module 02's reference. Everything about index construction and compression, from the people who measured it.
Sanderson — Test Collection Based Evaluation of IR Systems (free)	Module 05's history and its critique — how the evaluation methodology developed, and what it assumes that is not true.
The TREC proceedings, and BEIR (free)	The field's evaluation infrastructure. TREC for the methodology, BEIR for the zero-shot comparisons Module 07 relies on.

Module-by-module map

Module 01 — The Information Need

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapter 1 (free)	§1 and §2 , and the Boolean retrieval starting point that the rest of the course departs from.
Broder — A taxonomy of web search (free)	§3's taxonomy in the original — short, and the navigational/informational/transactional split still organises practice.
Saracevic — Relevance: A review of the literature (free)	§3's callout, developed at length — relevance as a relation, with the several decades of argument behind it.
Furnas et al. — The vocabulary problem (free)	§2 measured — how rarely two people choose the same word for the same thing, which is the quantitative basis for the whole module.

Module 02 — The Index

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapters 1 through 5 (free)	The whole module , with the construction algorithms and the compression codes derived.
Zobel & Moffat — Inverted files for text search engines (free)	§3 and §4 exhaustively — the reference, with the measurements behind every recommendation.
Lemire & Boytsov — Decoding billions of integers per second (free)	§3's modern schemes — why decode throughput beat compression ratio, with the numbers.
The Lucene index format documentation (free)	§1 through §4 as actually implemented — read the postings format description against this module.

Module 03 — Ranking

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapters 6, 11, and 12 (free)	§1 through §4 , with the probabilistic derivations done carefully.
Robertson & Zaragoza — The Probabilistic Relevance Framework (free)	§3 in the original , by its authors — including where BM25 came from and what it approximates.
Zhai & Lafferty — A study of smoothing methods (free)	§4's smoothing, measured — and the IDF-emerges-from-smoothing argument.
Thakur et al. — BEIR (free)	§3's callout, with the numbers — BM25 against neural methods across eighteen zero-shot collections.

Module 04 — Query Processing

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapter 7 (free)	§1 through §3 — the processing orders and the inexact top-k strategies.
Broder et al. — Efficient query evaluation using a two-level retrieval process (free)	§2's WAND in the original — short, and the bound argument is the whole paper.
Ding & Suel — Faster top-k document retrieval using block-max indexes (free)	§2's tightening — block-max WAND, which is what current engines implement.
Dean & Barroso — The Tail at Scale (free)	§4's fan-out problem — why partitioning worsens the tail, and what to do about it.

Module 05 — Evaluation

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapter 8 (free)	§1 and §3 — the metrics defined carefully, with the test collection methodology.
Sanderson — Test Collection Based Evaluation (free)	§1 and §2 critically — the assumptions, the pooling bias, and the judge agreement literature.
Chapelle et al. — Large-scale validation of click models (free)	§4's position bias, measured — and the models that correct for it.
Radlinski & Craswell — Comparing the sensitivity of information retrieval metrics (free)	§4's interleaving argument — how much more sensitive it is, with the numbers.

Module 06 — Learning to Rank

Resource	What it covers for this module
Liu — Learning to Rank for Information Retrieval (free survey)	§1's three formulations , organised, with the theoretical relationships between them.
Burges — From RankNet to LambdaRank to LambdaMART (free)	§2 by its author — the clearest account of the lambda trick and why it works.
Qin et al. — Are Neural Rankers still Outperformed by Gradient Boosted Decision Trees? (free)	§2's continued relevance, measured — on tabular ranking features, which is the setting that matters here.
Joachims et al. — Unbiased Learning-to-Rank with Biased Feedback (free)	§3's position bias problem, with the correction — inverse propensity weighting for click data.

Module 07 — Dense Retrieval

Resource	What it covers for this module
Lin, Nogueira & Yates — Pretrained Transformers for Text Ranking (free)	The whole module — and the most careful available assessment of which results replicated.
Karpukhin et al. — Dense Passage Retrieval (free)	§2's training in the original — including the in-batch negatives finding.
Thakur et al. — BEIR (free)	§3's out-of-domain result — the benchmark that showed how much less dense retrieval transfers.
Khattab & Zaharia — ColBERT (free)	§3's multi-vector answer — late interaction, and the index size it costs.

Module 08 — Vector Search at Scale

Resource	What it covers for this module
Malkov & Yashunin — HNSW (free)	§2's first family in the original — the layered navigable graph, and why greedy search on it works.
Jégou, Douze & Schmid — Product Quantization (free)	§3's main technique — the subspace decomposition and the lookup-table distance computation.
The FAISS documentation and wiki (free)	§4 practically — the index type guidance, which is the best available decision tree for this.
ANN-Benchmarks (free)	§2 and §4 measured — recall against latency across methods and datasets, which is how to compare them.

Module 09 — Query Understanding

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapters 3 and 9 (free)	§1's correction and §2 and §3's expansion and feedback , with Rocchio derived.
Croft, Metzler & Strohman, chapter 6 (free PDF)	§4's query understanding — segmentation, intent, and the production pipeline shape.
Cucerzan & Brill — Spelling correction as an iterative process (free)	§1's log-based approach — why the query log beats a dictionary, with the method.
Gao et al. — Precise Zero-Shot Dense Retrieval without Relevance Labels (free)	§4's callout — generating a hypothetical document to retrieve against, which is the clearest instance of the shift.

Module 10 — Web-Scale Retrieval

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapters 19 through 21 (free)	§1 through §3 — crawling, duplicate detection, and link analysis, with the algorithms.

Resource	What it covers for this module
Croft, Metzler & Strohman, chapters 3 and 9 (free PDF)	§1's mechanics and §4's spam material , in more practical detail.
Brin & Page — The Anatomy of a Large-Scale Hypertextual Web Search Engine (free)	§3 and §4 in the original — and worth reading for how much of the architecture survived.
Gyöngyi & Garcia-Molina — Web Spam Taxonomy (free)	§4's techniques, catalogued — the defensive equivalent of CSCE 701's threat modelling.

Module 11 — Generative Retrieval

Resource	What it covers for this module
Lewis et al. — Retrieval-Augmented Generation (free)	§1's architecture in the original — read it against CSCE 638 Module 11, which covers the generation side.
Rashkin et al. — Measuring Attribution in NLG (free)	§2's requirement, formalised — what it means for a citation to support a claim, and how to judge it.
Liu, Zhang & Liang — Evaluating Verifiability in Generative Search Engines (free)	§2 and §3 measured on deployed systems — and the attribution rates are sobering.
Nakano et al. — WebGPT (free)	§1's pipeline with retrieval in the loop — and the honest discussion of what the citations did and did not establish.

Module 12 — Exposure and Consequences

Resource	What it covers for this module
Singh & Joachims — Fairness of Exposure in Rankings (free)	§1 and §3 formalised — exposure as the allocated resource, with the optimisation framework.
Biega, Gummadi & Weikum — Equity of Attention (free)	§3's fourth row — amortised fairness, which is the most implementable of the criteria.
Joachims et al. — Unbiased Learning-to-Rank with Biased Feedback (free)	§2's correction — inverse propensity weighting, and what it requires to work.
Bakshy, Messing & Adamic, and the subsequent debate (free)	§4's counter-evidence and its critics — read the paper and at least one critique, since the disagreement is the lesson.

Module 13 — Claiming Relevance Honestly

Resource	What it covers for this module
Armstrong et al. — Improvements that don't add up (free)	§1's problem, measured across a decade of papers — weak baselines producing apparent cumulative progress that was not there.
Lin — The Neural Hype and Comparisons Against Weak Baselines (free)	§1's fourth row — the tuned-baseline argument, stated directly.
Fuhr — Some Common Mistakes In IR Evaluation (free)	§1 and §2 — a short, pointed list, and worth checking your own work against.
Sakai — Statistical Reform in Information Retrieval? (free)	§2's reporting requirements — significance testing, effect sizes, and the per-query distribution.